

SW-COD:

- Настройка имени:

```
hostnamectl set-hostname sw-cod.au.team; exec bash
```

- Настройка пользователя:
 - создаём пользователя

```
useradd net_admin
```

- задаём пароль пользователю

```
passwd net_admin
```

- должно получиться следующее

```
[root@sw-cod ~]# passwd net_admin
passwd: updating all authentication tokens for user net_admin.
```

You can now choose the new password or passphrase.

A valid password should be a mix of upper and lower case letters, digits, and other characters. You can use a password containing at least 7 characters from all of these classes, or a password containing at least 8 characters from just 3 of these 4 classes.

An upper case letter that begins the password and a digit that ends it do not count towards the number of character classes used.

A passphrase should be of at least 3 words, 11 to 72 characters long, and contain enough different characters.

Alternatively, if no one else can see your terminal now, you can pick this as your password: "Geneva+Clasp-Aside".

Enter new password:

Weak password: based on a dictionary word and not a passphrase.

Re-type new password:

```
passwd: all authentication tokens updated successfully.
```

- Добавляем пользователя в группу **wheel**:

```
usermod -aG wheel net_admin
```

- Настраиваем **sudo** для пользователя:

```
echo "net_admin ALL=(ALL:ALL) NOPASSWD: ALL" >> /etc/sudoers
```

- Проверить:

```
sw-cod login: net_admin
Password:
[net_admin@sw-cod ~]$ sudo hostname -f
sw-cod.au.team
[net_admin@sw-cod ~]$ exit
ВЫХОД
```

Welcome to ALT Server 11.0 (Mendelevium)!

- Настройка временного IP для установки openvswitch:

```
ip addr add 172.16.1.0/23 dev ens19
ip route add 0.0.0.0/0 via 172.16.1.254
echo "nameserver 77.88.8.8" > /etc/resolv.conf
```

- Устанавливаем пакет **openvswitch**:

```
apt-get update && apt-get install -y openvswitch
```

- Включаем и добавляем в автозагрузку службу **openvswitch**:
 - также реализуем сохранение настроек созданных средствами утилиты **ovs-vsctl**
 - и перезагружаем виртуальную машину

```
systemctl enable --now openvswitch
sed -i "s/OVS_REMOVE=yes/OVS_REMOVE=no/g" /etc/net/ifaces/default/options
reboot
```

- Настраиваем коммутацию:
 - включаем все существующие сетевые интерфейсы

```
echo "TYPE=eth" > /etc/net/ifaces/ens19/options
cp -r /etc/net/ifaces/ens{19,20}/
cp -r /etc/net/ifaces/ens{19,21}/
cp -r /etc/net/ifaces/ens{19,22}/
cp -r /etc/net/ifaces/ens19 /etc/net/ifaces/enp2s29
cp -r /etc/net/ifaces/ens19 /etc/net/ifaces/enp3s12
systemctl restart network
```

- [Проверить:](#)

```
[root@sw-cod ~]# ip -c -br a
lo                UNKNOWN        127.0.0.1/8 ::1/128
ens19             UP                fe80::be24:11ff:febf:d8fb/64
ens20             UP                fe80::be24:11ff:fe41:8167/64
ens21             UP                fe80::be24:11ff:fe0d:fa1f/64
ens22             UP                fe80::be24:11ff:fe76:cab8/64
enp2s29           UP                fe80::be24:11ff:fe7e:9a42/64
enp3s12           UP                fe80::be24:11ff:fe7d:469d/64
```

- [Создаём коммутатор с именем **sw-cod**:](#)

```
ovs-vsctl add-br sw-cod
```

- [Добавляем все интерфейсы в коммутатор **sw-cod**:](#)

```
ovs-vsctl add-port sw-cod ens19
ovs-vsctl add-port sw-cod ens20
ovs-vsctl add-port sw-cod ens21
ovs-vsctl add-port sw-cod ens22
ovs-vsctl add-port sw-cod enp2s29
ovs-vsctl add-port sw-cod enp3s12
```

- [Проверить:](#)

```
[root@sw-cod ~]# ovs-vsctl show
43b9c49d-4803-4df5-9b65-bd8efab074d0
    Bridge sw-cod
        Port enp2s29
            Interface enp2s29
        Port sw-cod
            Interface sw-cod
                type: internal
        Port ens19
            Interface ens19
        Port ens21
            Interface ens21
        Port ens22
            Interface ens22
        Port enp3s12
            Interface enp3s12
        Port ens20
            Interface ens20
    ovs_version: "3.3.8"
[root@sw-cod ~]#
```

- [Создаём порт управления для назначения IP-адреса:](#)

```
mkdir /etc/net/ifaces/mgmt

cat <<EOF > /etc/net/ifaces/mgmt/options
TYPE=ovsport
BOOTPROTO=static
CONFIG_IPv4=yes
BRIDGE=sw-cod
EOF

echo "172.16.1.0/23" > /etc/net/ifaces/mgmt/ipv4address
echo "default via 172.16.1.254" > /etc/net/ifaces/mgmt/ipv4route
echo "search au.team" > /etc/net/ifaces/mgmt/resolv.conf
echo "nameserver 10.1.1.10" >> /etc/net/ifaces/mgmt/resolv.conf
systemctl restart network
```

- [Проверить:](#)

```
[root@sw-cod ~]# ip -c -br -4 a
lo                UNKNOWN          127.0.0.1/8
mgmt              UNKNOWN          172.16.1.0/23
[root@sw-cod ~]# ip -c r
default via 172.16.1.254 dev mgmt
172.16.0.0/23 dev mgmt proto kernel scope link src 172.16.1.0
[root@sw-cod ~]# cat /etc/resolv.conf
# Generated by resolvconf
# Do not edit manually, use
# /etc/net/ifaces/<interface>/resolv.conf instead.
search au.team
nameserver 10.1.1.10
[root@sw-cod ~]# ovs-vsctl show
43b9c49d-4803-4df5-9b65-bd8efab074d0
    Bridge sw-cod
        Port enp2s29
            Interface enp2s29
        Port sw-cod
            Interface sw-cod
                type: internal
        Port ens19
            Interface ens19
        Port mgmt
            Interface mgmt
                type: internal
        Port ens21
            Interface ens21
        Port ens22
            Interface ens22
        Port enp3s12
            Interface enp3s12
        Port ens20
            Interface ens20
    ovs_version: "3.3.8"
[root@sw-cod ~]# ping -c3 77.88.8.8
PING 77.88.8.8 (77.88.8.8) 56(84) bytes of data.
64 bytes from 77.88.8.8: icmp_seq=1 ttl=54 time=71.8 ms
64 bytes from 77.88.8.8: icmp_seq=2 ttl=54 time=70.3 ms
64 bytes from 77.88.8.8: icmp_seq=3 ttl=54 time=68.6 ms

--- 77.88.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 68.591/70.221/71.815/1.316 ms
[root@sw-cod ~]#
```